Conv-J2SMamba: A New Hybrid Convolutional and Joint Scan Merge Structured Mamba Network for Hyperspectral Image Classification

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***Abstract*—Recently, Mamba has shown great potential in long-sequence modeling and has begun to be applied to hyperspectral image classification (HSIC). However, Mamba exhibits relatively weak local feature modeling capability. Moreover, as a causal sequence model, Mamba can only obtain a global receptive field at the end of the sequence, which fundamentally limits its ability to capture the complex spatial structures of hyperspectral images (HSIs). To address this problem, this article proposes a new hybrid convolution and joint scan merge structured Mamba network (Conv-J2SMamba) for HSIC, designed to capture the complex spatial structures and local–global features of HSIs. Specifically, we employ a convolutional branch to extract local features and design a joint scan merge structured Mamba (J2SMamba) to deeply model HSI spatial structures and capture global contextual features. Within J2SMamba, a joint scan merge (JSM) strategy and a structure-aware state-space model (SASSM) are constructed. The JSM strategy concatenates and integrates dual HSIs, followed by joint scan and joint merge operations. Compared with conventional scanning methods, this strategy enables each pixel location to obtain a global omnidirectional receptive field while maintaining the original sequence length unchanged. SASSM further employs the structure-aware state fusion (SASF) equation to capture spatial structural dependencies in HSIs, significantly enhancing the flow and propagation of contextual information. To further integrate the generated features, a contrastive-guided feature fusion (CGFF) module is also introduced, which utilizes a multilevel attention mechanism to guide the key feature enhancement and multilevel feature fusion. Experimental results on four public HSIs datasets demonstrate that the proposed Conv-J2SMamba model outperforms state-of-the-art methods.**

***Index Terms*—Contrastive-guided feature fusion (CGFF) module, hyperspectral image classification (HSIC), joint scan merge (JSM) strategy, structure-aware state-space model (SASSM).**

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I. Introduction

HYPERSPECTRAL images (HSIs) are acquired by satellite sensors that measure solar radiation reflected from the Earth's surface, capturing rich spectral information across multiple contiguous bands [1], [2], [3], [4]. Compared with RGB or multispectral images, HSIs capture a large amount of fine-grained spectral information over continuous and narrow spectral bands, providing stronger discriminative capability for the precise identification of surface materials [5]. As a result, HSIs have been widely applied to precision agriculture [6], ecological protection [7], quality inspection [8], mineral resource exploration [9], and other fields. Hyperspectral image classification (HSIC), as a crucial step in HSI processing and analysis, aims to assign a class label to each pixel in the remotely sensed imagery [10].

A. Traditional HSIC Methods

Traditional HSIC methods mainly include handcrafted approaches and machine-learning-based techniques, such as extreme learning machine (ELM) [11], support vector machine (SVM) [12], random forest [13], principal component analysis (PCA) [14], and maximum likelihood classification (MLC) [15]. To address the challenges posed by the rich spatial contextual details contained in HSIs, methods such as extended morphological profiles (EMP) [16], superpixel segmentation [17], and extended multiattribute profiles (EMAPs) [18] have been introduced into HSIC. However, these methods, as shallow learning models, heavily rely on prior information obtained during training, and their representation capability is limited under complex environmental conditions.

B. Deep-Learning-Based HSIC Methods

With the powerful nonlinear representation capacity of deep learning [19], [20], [21], [22], HSIC has made remarkable advancements. For example, Hu et al. [23] were the first

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to propose a 1-D convolutional neural network (CNN) for extracting spectrally discriminative features. Considering the spatial dependencies in HSIs, He et al. [24] constructed a multiscale 3-D CNN, which is capable of learning spectral and spatial features in an end-to-end manner. Wang et al. [25] introduced dense connections, which allowed the model to fully utilize the characteristics among different convolutional layers and achieve better classificatory performance. Li et al. [26] developed a salient object extraction method based on the multiscale region growing search, which separately extracts global and local features. To address the issue of mixed pixels (i.e., pixels made up of several spectral constituents), Han et al. [27] presented a subpixel-guided network that automatically integrates convolutional features with subpixel information. Despite their superior performance over traditional algorithms, these methods still struggle to capture spatial–contextual dependencies in HSIs because of the limited receptive field of convolutional kernels.

1) Transformer-Based HSIC Methods: Benefiting from the exceptional long-range modeling capability of attention mechanisms [28], [29], [30], transformers are frequently employed for global feature extraction [31]. Hong et al. [32] introduced SpectralFormer to capture subtle variations among spectral and global characteristics. Roy et al. [33] developed the Morphological Attention Transformer, which enhances the structural information interaction between HSI and CLS (classification) tokens by integrating the attention mechanism with learnable spectral–spatial morphological convolutions. Jia et al. [34] innovatively introduced a regularized central mask-based pretraining task, enabling Transformers to model the dependencies between central pixels and neighboring classes for the first time, which significantly improved HSIC performance. Sun et al. [35] presented a memory-enhanced spatial–spectral transformer architecture, which utilizes a memory mechanism to store spectral–spatial features and enhance self-attention, effectively mitigating the information loss and gradient vanishing in deep models. Although Transformers excel in global modeling, the quadratic computational complexity [36] of the self-attention mechanism poses a challenge to computational efficiency.

2) Mamba-Based HSIC Methods: Recently, the Mamba model—constructed upon state-space models (SSMs)—has demonstrated significant potential in HSIC, as it can perform long-range sequence modeling while preserving the linear computational complexity [37], [38]. Gu et al. [39] first revealed Mamba's powerful global context modeling capability. Li et al. [40] were the first to introduce Mamba into HSIC, achieving performance that surpassed that of Transformers while maintaining linear complexity. He et al. [41] presented a spatial–spectral token generation mechanism and a 3-D selective scanning mechanism, successfully applying Mamba in high-dimensional spaces and enabling global modeling of HSI. Wang et al. [42] introduced a patchwise cross-scan strategy to capture spatial features, employed bidirectional spectral scanning to explore spectral characteristics, and used a gating controller to fuse spatial–spectral features adaptively. To address the problem of information redundancy and enhance computational efficiency, He et al. [43] introduced a lightweight interval-grouped scanning method, significantly reducing the number of parameters. However, despite Mamba's impressive performance in terms of computational complexity, it still suffers from insufficient representational capacity when modeling the complex spatial structures of HSIs [44]. Therefore, it is crucial to explore a structured Mamba.

C. Our Contribution

Based on the above-mentioned problem, this article introduces a hybrid convolution and joint scan-merge structured Mamba network (Conv-J2SMamba) for HSIC. By establishing neighborhood connectivity within the SSM, the proposed network effectively captures the complex spatial structures of HSIs while simultaneously extracting both local and global features. Since HSIs contain abundant local texture information as well as global contextual information, whereas Mamba exhibits relatively limited capability in modeling local features, Conv-J2SMamba adopts a dual-branch architecture. Specifically, the convolutional branch focuses on fine-grained local feature extraction, while the joint scan merge structured Mamba (J2SMamba) branch is responsible for modeling global contextual dependencies. The J2SMamba branch consists of two core components: the joint scan merge (JSM) mechanism and the structure-aware SSM (SASSM). To address the challenges posed by the complex spatial structures of HSIs and the difficulty of conventional scanning methods in simultaneously achieving the computational efficiency and omnidirectional global receptive fields, this article designs a JSM mechanism. This mechanism enhances interpixel information interaction by concatenating dual-HSI images and constructs stride-based four-directional scanning paths for selective scanning without increasing the overall sequence length. Subsequently, omnidirectional global features are aggregated through the joint merge operation. Building upon this, the SASSM introduces a structure-aware state fusion (SASF) equation, which explicitly injects the spatial neighborhood structural dependencies into the state transition process. This enhances the flow of contextual information within the SSM and enables deep modeling of HSI spatial structures. Finally, to fully integrate the local and global features extracted from the two branches, a contrastive-guided feature fusion (CGFF) module is proposed. Guided by fusion weights generated through a multilevel attention mechanism, CGFF adaptively fuses local features, global features, and initial HSIs features, thereby obtaining the most discriminative multilevel feature representations.

The main innovative contributions of our work can be summarized as follows.

1) A new Conv-J2SMamba network is developed for HSIC, which establishes neighborhood connections within the state-space model to capture the complex spatial structures of HSIs while simultaneously extracting both global and local features.
2) A restructured JSM mechanism is introduced to comprehensively represent HSI features without increasing computational complexity. Meanwhile, the construction

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of SASSM significantly enhances the flow of contextual information within the SSM, thereby effectively improving the capability of capturing complex spatial structures in HSI.

3) A CGFF module based on multilevel attention-generated fusion weights is designed to guide the effective fusion of local, global, and initial HSI features, thereby further enhancing the model's cross-level feature representation capability.

D. Paper Organization

The remainder of this article is organized as follows. Section II presents an overview of related algorithms. The proposed approach is detailed in Section III. Section IV describes the experimental results obtained on four benchmark HSI datasets. Section V concludes this article with some remarks and hints at plausible future research directions.

II. Related Works

This section provides a brief overview of algorithms related to the SSM and several classic scanning methods. We begin with an introduction to the SSM, followed by the incorporation of selective mechanisms to form the Selective Structured SSM (S6). Then, we focus on the classic scanning algorithms used in VMamba and EVMamba, laying the groundwork for the proposed scanning strategy.

A. State-Space Model

The SSM is a mathematical framework used to describe the dynamic behavior of systems as they evolve over time [45]. Specifically, given an input $x(t) \in \mathbb{R}$, the sequence is mapped to an output $z(t) \in \mathbb{R}$ through a hidden state $s(t) \in \mathbb{R}^N$, enabling sequence modeling via the internal dynamics of the hidden state. The entire process can be described by the following equations:

$$\begin{aligned} s'(t) &= \mathbf{P}s(t) + \mathbf{Q}x(t) \\ z(t) &= \mathbf{W}s(t) \end{aligned} \tag{1}$$

where $\mathbf{P} \in \mathbb{R}^{N\times N}$ represents the state transition matrix, $\mathbf{Q} \in \mathbb{R}^{N\times 1}$ is the control input matrix, and $\mathbf{W} \in \mathbb{R}^{N\times 1}$ denotes the output matrix. These two equations characterize the system's state evolution and observation processes, respectively.

Given that the data in HSIC are typically discrete, the continuous parameters $\mathbf{P}$ and $\mathbf{Q}$ in the SSM are transformed into their discrete counterparts $\overline{\mathbf{P}}$ and $\overline{\mathbf{Q}}$ by introducing a discretization parameter Δ [46]. The resulting discrete system can be expressed as follows:

$$\begin{aligned} \overline{\mathbf{P}} &= \exp(\Delta\mathbf{P}) \\ \overline{\mathbf{Q}} &= (\Delta\mathbf{P})^{-1}\left(\exp(\Delta\mathbf{P}) - \mathbf{I}\right)\cdot\Delta\mathbf{Q}. \end{aligned} \tag{2}$$

After discretizing $\overline{\mathbf{P}}$ and $\overline{\mathbf{Q}}$, convolutional operations can accelerate the process, thereby enhancing the scalability and efficiency [47]. The final output can be represented as follows:

$$\begin{aligned} s_t &= \overline{\mathbf{P}}s_{t-1} + \overline{\mathbf{Q}}x(t) \\ z(t) &= \mathbf{W}s(t) \end{aligned}$$

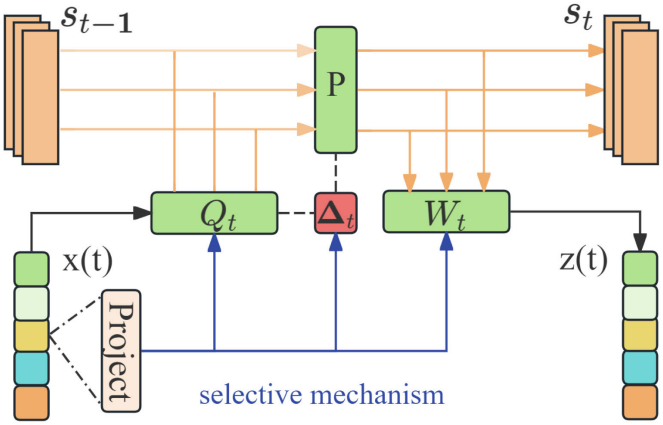


Fig. 1. Diagram illustrating selective structured SSMs (S6).

$$\begin{aligned} \overline{\mathbf{K}} &= \left(\mathbf{W}\overline{\mathbf{Q}}, \mathbf{W}\overline{\mathbf{P}}\overline{\mathbf{Q}}, \ldots, \mathbf{W}\overline{\mathbf{P}}^{\mathbf{L}-1}\overline{\mathbf{Q}}\right) \\ \mathbf{z} &= \mathbf{x} * \overline{\mathbf{K}} \end{aligned} \tag{3}$$

where $\mathbf{L}$ denotes the size of the input sequence x and $\overline{\mathbf{K}} \in \mathbb{R}^L$ corresponds to the structured convolutional kernel.

Traditional SSMs face difficulties in capturing contextual dependencies within sequences. To overcome this limitation, a selective mechanism was introduced, leading to the construction of the selective structured SSMs (S6) [48]. The structure of S6 is illustrated in Fig. 1. In S6, the input is parameterized by a selective mechanism, enabling the model to selectively focus on different parts of the sequence. Specifically, given the input $x \in \mathbb{R}^{B\times L\times D}$, the projection matrix can be formulated as follows:

$$\begin{aligned} \mathbf{Q} &= \text{Projection}_Q(x) \\ \mathbf{W} &= \text{Projection}_W(x) \\ \Delta &= \tau_\Delta\left(\text{Parameter} + \text{Projection}_\Delta(x)\right) \end{aligned} \tag{4}$$

where Projection_X denotes the projection onto an M-dimensional space, and τ_Δ applies the Softplus activation function. Here, $\mathbf{Q} \in \mathbb{R}^{B\times L\times M}$, $\mathbf{W} \in \mathbb{R}^{B\times L\times M}$, and $\Delta \in \mathbb{R}^{B\times L\times D}$. The selective mechanism enables the model to selectively process information, effectively filtering out irrelevant noise while preserving specific input features.

B. Scanning Strategies

Mamba has achieved outstanding performance in processing 1-D sequential data, including natural language processing (NLP) tasks. However, due to the unidirectional nature of Mamba and the nonsequential characteristics of visual data, it faces challenges when applied to vision tasks [49]. To model the complex spatial structures in visual data, Liu et al. [50] proposed VMamba, which employs 2-D selective scanning (SS2D) as its core module to establish a global receptive field for each pixel. In order to further reduce computational complexity, Pei et al. [51] introduced EfficientVMamba (EVMamba), adopting efficient 2-D scanning (ES2D) to improve computational efficiency while capturing global features. The main principles of these two methods are introduced below.

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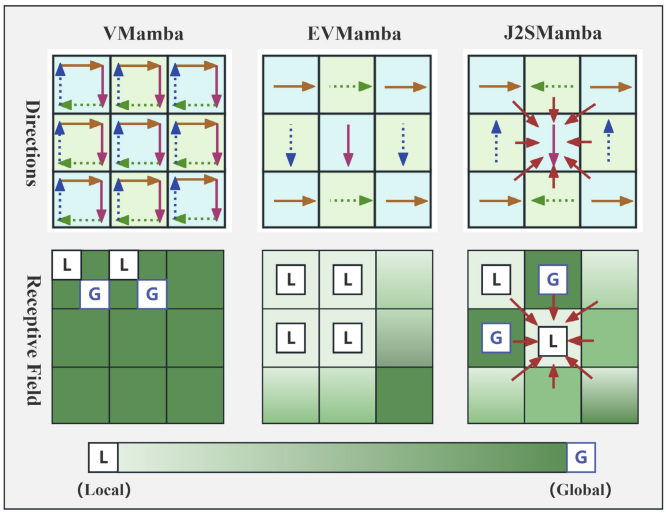


Fig. 2. Graphical illustration of the sequence scanning paths and receptive fields of different mamba models.

1) SS2D: The data processing flow of SS2D is divided into three stages: the cross-scanning stage, the selective processing stage based on S6 blocks, and the cross-fusion stage. Specifically, the input patches are first unfolded into sequences along four different directional paths. Then, for each sequence, independent S6 modules are applied in parallel. After processing, the sequences are rearranged and fused into the final output. Fig. 2 provides a schematic of the SS2D scanning method and its receptive field. It can be observed that SS2D allows every pixel to access a global receptive field. However, this scanning method increases the sequence length by four times, thereby adding to the computational burden significantly.

2) ES2D: Inspired by the use of strided convolutions for efficient sampling [52], ES2D introduces strided sampling by employing scanning blocks with a stride of p to skip pixels on the feature map. This effectively reduces the sequence length without degrading spatial resolution. The scanning paths and receptive fields of ES2D are depicted in Fig. 2. In contrast to SS2D, ES2D maintains the sequence length of the input data and improves the computational efficiency. Nevertheless, ES2D restricts the receptive field to the lower right region, resulting in an unbalanced receptive field distribution.

III. Proposed Method

This section first provides an overview of the overall architecture of our Conv-J2SMamba framework designed for HSIC, followed by a comprehensive description of its submodules. Specifically, a novel JSM Strategy is presented to perform more comprehensive and omnidirectional scanning of HSI data. After the scanning process, an SASF equation is introduced based on the S6 stage to form the SASSM module, enabling selective scanning of HSIs. Finally, the CGFF module is introduced to fuse the generated features.

A. Overall Architecture: Overview

Fig. 3 illustrates the architecture of the proposed Conv-J2SMamba model, which consists of four components. Specifically, the framework includes a PCA layer, a patch embedding layer, the Conv-J2SMamba block, and a classifier.

We first employ PCA to reduce the dimensionality of the input features, eliminating redundant information while preserving key characteristics. Then, a patch embedding layer projects the spectral vectors into the embedding space. The process of obtaining pixel embeddings $\mathbf{F} \in \mathbb{R}^{H\times W\times C}$ from the input features $\mathbf{I} \in \mathbb{R}^{H\times W\times D}$ can be described as follows:

$$\mathbf{F} = \text{PatchEmbedding}\left(\text{PCA}\left(\mathbf{I}\right)\right). \tag{5}$$

Then, the embedded features are fed into the Conv-J2SMamba blocks for both global and local feature extraction. Specifically, the convolutional branch extracts local features $\mathbf{F}_l \in \mathbb{R}^{H\times W\times C}$, while the J2SMamba branch is responsible for extracting global features $\mathbf{F}_g \in \mathbb{R}^{H\times W\times C}$. The CGFF module then performs multilevel feature fusion of $\mathbf{F}_g$, $\mathbf{F}_l$, and the initial feature $\mathbf{F}$. This procedure is formally defined as follows:

$$\begin{aligned}\mathbf{F}_l &= \text{Conv branch}\left(\mathbf{F}\right)\\ \mathbf{F}_g &= \text{J2SMamba branch}\left(\mathbf{F}\right)\\ \mathbf{F}_f &= \text{CGFF}\left(\mathbf{F}, \mathbf{F}_g, \mathbf{F}_l\right)\end{aligned} \tag{6}$$

where $\mathbf{F}_f \in \mathbb{R}^{H\times W\times C}$ denotes the fused features. Finally, the fused features are fed into the classifier to produce the class prediction for each pixel.

B. JSM Strategy

1) Joint Scan: Existing scanning methods are designed for single inputs, which may lead to feature loss. To address this, we concatenate and integrate identical inputs and then perform repeated sampling to enhance the model's generalization ability. The features $\mathbf{F} \in \mathbb{R}^{H\times W\times C}$ are first concatenated along the vertical and horizontal directions as follows:

$$\mathbf{X}^h = [\mathbf{F} \mid \mathbf{F}], \quad \mathbf{X}^w = \begin{bmatrix}\mathbf{F}\\ \mathbf{F}\end{bmatrix} \tag{7}$$

where $\mathbf{X}^h \in \mathbb{R}^{2H\times W\times C}$ refers to the features concatenated along the horizontal direction and $\mathbf{X}^w \in \mathbb{R}^{H\times 2W\times C}$ refers to the features concatenated along the vertical direction.

Bearing in mind that VMamba [50] increases the input sequence length from N to $4N$ (despite having a global receptive field), and that EVMamba [51] only supports forward scanning, resulting in an insufficient receptive field (while keeping the input sequence length as N), we combine the stride scanning strategy in EVMamba with the four-direction scanning strategy in VMamba to form a stride-based four-direction scanning strategy. Each feature obtains both global and omnidirectional receptive fields, while the input sequence length remains as N. The schematic of this scanning strategy is shown in Fig. 2.

After performing horizontal and vertical scanning on $\mathbf{X}^h$ and $\mathbf{X}^w$, four directional sequences $\{I_i\}_{i=1}^4$, corresponding to the right, left, up, and down directions, are generated. Each sequence has a length $N/4$. These sequences are then fed into Mamba blocks for selective scanning to establish long-range dependencies. The scanning process for the four directional sequences can be formulated as follows:

$$I_1 \overset{\text{horizontal}}{\Longleftarrow} \mathbf{X}^h[m :: s, n :: s, :]$$

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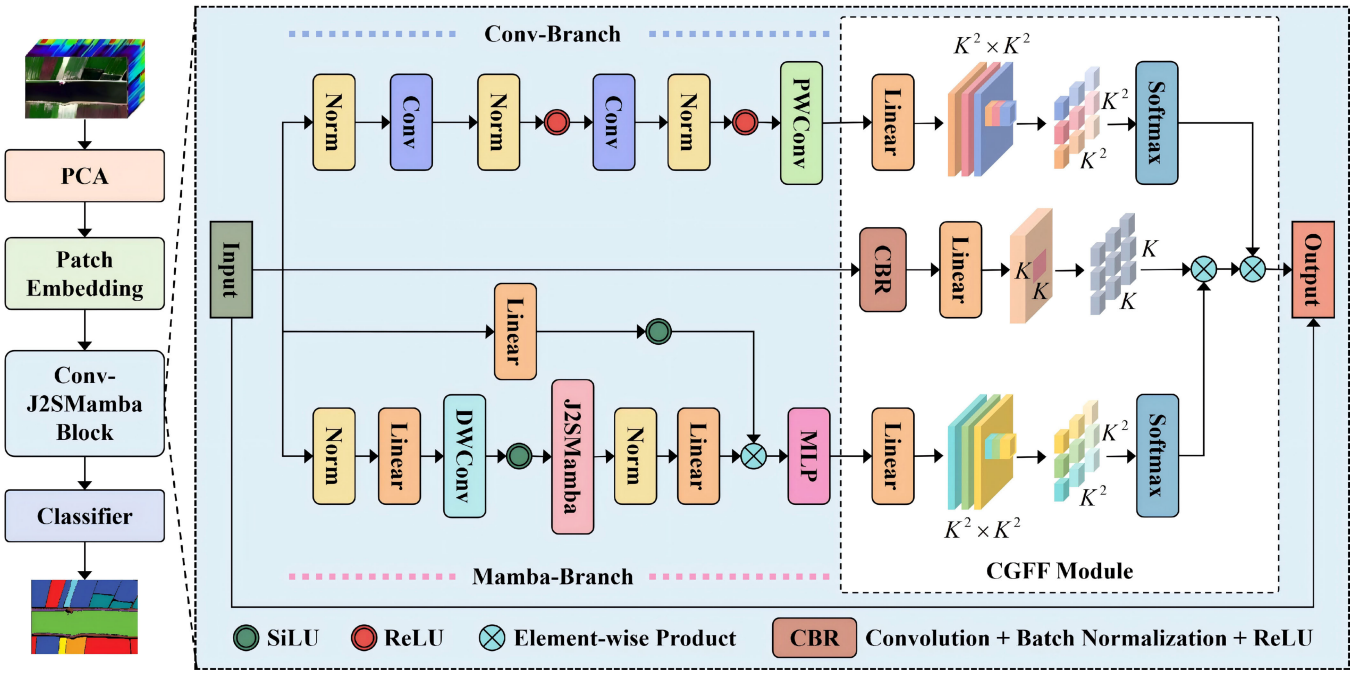


Fig. 3. Schematic of the proposed Conv-J2SMamba module. First, the HSI is fed into the PCA module for dimensionality reduction to remove redundant features, and the embedding layer projects the spectral vectors into an embedding space. Then, the features are passed through the Conv-J2SMamba block to extract both global and local representations, where the convolutional branch captures local information and the mamba branch models global dependencies. Finally, the CGFF module adaptively performs weighted fusion of global features, local features, and the initial HSI features.

$$
\begin{aligned}
I_2 &\overset{\text{horizontal}}{\Longleftarrow} \text{Flip}\left(\mathbf{X}^h\left[m::s, n::s, :\right]\right)\\
I_3 &\overset{\text{vertical}}{\Longleftarrow} \mathbf{X}^w\left[m::s, n::s, :\right]\\
I_4 &\overset{\text{vertical}}{\Longleftarrow} \text{Flip}\left(\mathbf{X}^w\left[m::s, n::s, :\right]\right)\\
\left\{\tilde{I}_i\right\}_{i=1}^4 &= \text{Mamba}\left(\left\{I_i\right\}_{i=1}^4\right)
\end{aligned} \tag{8}
$$

where $\{\tilde{I}_i\}_{i=1}^4$ is the sequence after the Mamba scanning. $[m::s, n::s, :]$ denotes the starting point for slicing the matrix of each channel from m and n. s represents the stride for the scanning process, which is set to 2 in our experiments. Flip indicates a reversal of the scanning direction. The starting points for the slicing (m, n) can be defined by the following formula:

$$
\begin{cases}
m = \left\lfloor \dfrac{i-1}{2} \right\rfloor, \\
n = (i-1) \bmod 2,
\end{cases}
\qquad i \in \{1, 2, 3, 4\}. \tag{9}
$$

2) Joint Merge: After building long-range dependencies for the sequences in each direction, reintegrating them into the feature map becomes a challenge. To tackle this, we propose a Joint Merge strategy, which reconstructs the features from the four scanned sequences back to their original spatial positions. The procedure can be mathematically expressed as follows:

$$
\begin{aligned}
l\tilde{I}_{2,4} &= \text{Flip}\left(\tilde{I}_{2,4}\right)\\
\mathbf{Y}^h\left[m::s, n::s, :\right] &\overset{\text{restore}}{\Longleftarrow} \tilde{I}_{1,2}\\
\mathbf{Y}^w\left[m::s, n::s, :\right] &\overset{\text{restore}}{\Longleftarrow} \tilde{I}_{3,4}
\end{aligned} \tag{10}
$$

where $\mathbf{Y}^h \in \mathbb{R}^{2H\times W\times C}$ and $\mathbf{Y}^w \in \mathbb{R}^{H\times 2W\times C}$ are the restored horizontal and vertical features, respectively, and $\mathbf{Y}^h$ and $\mathbf{Y}^w$ are divided along the horizontal and vertical directions, respectively, and then, the horizontal and vertical features are summed to obtain the fused feature $\tilde{\mathbf{F}} \in \mathbb{R}^{H\times W\times C}$

$$
\begin{aligned}
\mathbf{F_1} &= \mathbf{Y}_1^h + \mathbf{Y}_1^w = \mathbf{Y}_h\left[:H, :, :\right] + \mathbf{Y}_w\left[:, :W, :\right]\\
\mathbf{F_2} &= \mathbf{Y}_2^h + \mathbf{Y}_2^w = \mathbf{Y}_h\left[H:, :, :\right] + \mathbf{Y}_w\left[:, W:, :\right]\\
\tilde{\mathbf{F}} &= \mathbf{F_1} + \mathbf{F_2}.
\end{aligned} \tag{11}
$$

At the microscopic level, each feature still receives unidirectional information, resulting in a local receptive field. Therefore, we employ convolution as an aggregator to fuse information from different directions

$$
\mathbf{F}_g = \text{Conv}\left(\tilde{\mathbf{F}}\right) \tag{12}
$$

where Conv denotes a convolution with a kernel size of 3. $\mathbf{F}_g$ represents the final obtained global feature. This operation ensures that the resulting features are omnidirectional at both the microscopic and macroscopic levels, achieving a global receptive field. This significantly enhances the performance of Mamba.

C. Structure-Aware SSM

SSMs demonstrate strong capability in capturing long-range correlations within 1-D sequential data, but when applied to HSIC, they often disrupt the spatial structure of the HSI [53]. To capture the complex spatial structures of adjacent features in the data, we introduce a SASF equation into the SSM, resulting in an SASSM. The complete computational procedure of SASSM can be mathematically expressed through these equations

$$
x_t = \overline{\mathbf{P}}_t x_{t-1} + \overline{\mathbf{Q}}_t u_t
$$

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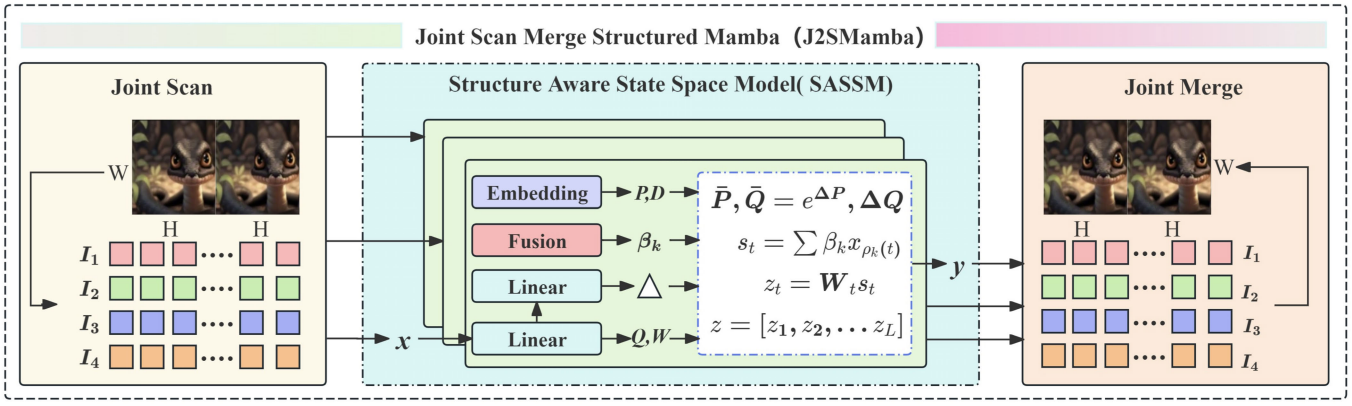


Fig. 4. Algorithmic flowchart of the proposed J2SMamba. First, two HSIs are concatenated, followed by the construction of four scanning paths. Then, the data are fed into SASSM for selective scanning and, finally, the original spatial structure is restored.

$$s_t = \sum_{k\in\Omega} \beta_k x_{\rho_k(t)}$$
$$z_t = \mathbf{W}_t s_t \tag{13}$$

where the symbol s_t represents the structure-aware state variable and β_k is a learnable weight. Ω refers to the neighborhood set, and $\rho_k(t)$ denotes the kth index at position t.

Fig. 4 provides an algorithmic flowchart of SASSM, which consists of three main stages. First, the input data are transformed into a 1-D sequence of length N using the previously described joint scan strategy. Next, the state representations are obtained through the state transition equation. Finally, the joint merge strategy is applied to reconstruct the original image structure. For the updated state s_t, a weighted linear combination of its neighboring states $\rho_k(t)$ within the predefined set Ω is computed, where the weight β_k determines the contribution of each neighbor to the final representation, thus obtaining representations with more contextual features.

Our method employs dilated convolutions, integrating local information into the new state to construct spatial relationship representations. Specifically, filters with dilation factors $d = 1, 3, 5$ are used to form the neighborhood sets Ω_d as follows:

$$s_t = \sum_{d=1,3,5} \sum_{m,n\in\Omega_d} k^d_{mn} \cdot x_{t+mw+n} \tag{14}$$

where k^d_{mn} represents the weight of the filter with dilation factor d at position (m, n), w stands for the image width, and x_{t+mw+n} corresponds to the neighbor of s_t located at (m, n). This strategy exploits the spatial dependencies within the HSI, leading to more precise and context-sensitive representations.

D. Contrastive-Guided Feature Fusion (CGFF)

The CGFF module is designed to facilitate multilevel feature fusion among global, local, and initial features. It helps the model integrate informative features while filtering out irrelevant ones. At any spatial coordinate (i, j), the CGFF module derives the attention weights over the $K \times K$ receptive field centered at that point by aggregating multilevel feature representations.

Specifically, given the input feature map $\mathbf{F} \in \mathbb{R}^{H\times W\times C}$, initial feature integration is performed using a CBR module, where CBR represents convolution plus batch normalization plus ReLU activation function. Then, the C-dimensional feature vector at each spatial position is projected into a value vector $\mathbf{T} \in \mathbb{R}^{H\times W\times C}$ via a linear transformation with weights $\mathbf{S}_T \in \mathbb{R}^{C\times C}$. These value vectors are subsequently unfolded over local windows in preparation for neighborhood feature aggregation. The matrix $\mathbf{T}_{\Delta_{i,j}} \in \mathbb{R}^{C\times K^2}$, which consists of all value vectors within the local region centered at position (i, j), is constructed as follows:

$$\mathbf{T}_{\Delta_{i,j}} = \left\{\mathbf{T}_{i+a-\left[\frac{K}{2}\right],j+b-\left[\frac{K}{2}\right]}\right\}, \quad 0 \le a, \quad b < K. \tag{15}$$

Then, the attention weights for global and local information are generated. $\mathbf{F}_g \in \mathbb{R}^{H\times W\times C}$ denotes the global features, $\mathbf{F}_l \in \mathbb{R}^{H\times W\times C}$ represents the local features, and $\mathbf{S}_g \in \mathbb{R}^{C\times K^4}$ and $\mathbf{S}_l \in \mathbb{R}^{C\times K^4}$ are the weight matrices obtained through linear transformations of the corresponding features. The attention weight is computed as follows:

$$\mathbf{E}_g = \mathbf{S}_g \cdot \mathbf{F}_g, \quad \mathbf{E}_f = \mathbf{S}_l \cdot \mathbf{F}_l \tag{16}$$

where $\mathbf{E}_g \in \mathbb{R}^{H\times W\times K^2}$ and $\mathbf{E}_f \in \mathbb{R}^{H\times W\times K^2}$ represent the attention weights corresponding to features $\mathbf{F}_g$ and $\mathbf{F}_l$, respectively. These weights at position (i, j) are then reshaped into $\hat{\mathbf{E}}_{g_{i,j}}$ and $\hat{\mathbf{E}}_{l_{i,j}} \in \mathbb{R}^{K^2\times K^2}$. Finally, the unfolded value vector $\mathbf{T}_{\Delta_{i,j}}$ is fused through a two-step weighted aggregation process

$$\tilde{\mathbf{T}}_{\Delta_{i,j}} = \text{Softmax}\left(\hat{\mathbf{E}}_{l_{i,j}}\right) \otimes \left(\text{Softmax}\left(\hat{\mathbf{E}}_{g_{i,j}}\right) \otimes \mathbf{T}_{\Delta_{i,j}}\right) \tag{17}$$

where $\otimes$ represents the matrix multiplication. Finally, the weighted values are subsequently fused through dense aggregation to generate the output feature map. Through this method, the original, local, and global features are fused in a weighted manner to produce the most representative output features.

IV. Experiments

A. Datasets

This section introduces four datasets employed for model performance assessment: University of Pavia (UP), Salinas Valley (SV), Fanglu Tea Farm (CY), and WHU-Hi-LongKou (LongKou).

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TABLE I
CLASS NUMBERS AND COLORS IN THE UNIVERSITY OF PAVIA DATASET

No.	Color	Class Name	Training	Test	Total
1		Asphalt	132	6499	6631
2		Meadows	372	18,277	18,649
3		Gravel	41	2058	2099
4		Trees	61	3003	3064
5		Mental sheets	26	1319	1345
6		Bare soil	100	4929	5029
7		Bitumen	26	1304	1330
8		Bricks	73	3609	3682
9		Shadow	18	929	947
		Total	855	41921	42776

TABLE II
CLASS NUMBERS AND COLORS IN THE SALINAS VALLEY DATASET

No.	Color	Class Name	Training	Test	Total
1		Brocoli-green-weeds-1	19	1950	1969
2		Brocoli-green-weeds-2	36	3,616	3,652
3		Fallow	19	1919	1938
4		Fallow-rough-plow	13	1355	1368
5		Fallow-smooth	26	2600	2626
6		Stubble	38	3843	3881
7		Celery	35	3474	3509
8		Grapes-untrained	110	10937	11047
9		Soil-vinyard-develop	60	6019	6079
10		Corn-senesced-green-weeds	32	3182	3214
11		Lettuce-romaine-4wk	10	1038	1048
12		Lettuce-romaine-5wk	18	1871	1889
13		Lettuce-romaine-6wk	8	890	898
14		Lettuce-romaine-7wk	10	1040	1050
15		Vinyard-untrained	71	1898	7124
16		Vinyard-vertical-trellis	17	1754	1771
		Total	522	52541	53063

TABLE III
CLASS NUMBERS AND COLORS IN THE FANGLU TEA FARM DATASET

No.	Color	Class Name	Training	Test	Total
1		Masson pine	116	5690	5806
2		Bamboo forest	46	2,262	2,308
3		Tea tree	568	27860	28428
4		Reed	4	210	214
5		Rice	136	6673	6809
6		Sweet potato	16	801	817
7		Coriander	8	421	429
8		Waste grass	37	1824	1861
9		Ponds	18	893	911
10		Building/road	18	929	947
		Total	1074	52660	53734

TABLE IV
CLASS NUMBERS AND COLORS IN THE WHU-HI-LONGKOU DATASET

No.	Color	Class Name	Training	Test	Total
1		Corn	345	34166	34511
2		Cotton	83	8,291	8,374
3		Sesame	30	3001	3031
4		Broad-leaf soybean	632	62580	63212
5		Narrow-leaf soybean	41	4110	4151
6		Rice	118	11736	11854
7		Water	670	66386	67056
8		Roads and houses	71	7053	7124
9		Mixed weed	52	5177	5229
		Total	2042	202500	204542

1) University of Pavia: The UP is an HSI dataset captured by the Reflective Optics Spectrographic Imaging System (ROSIS) sensor over Pavia, Italy. The image contains 103 spectral bands and measures 610 × 340 pixels in size. It consists of nine classes, totaling 42 776 annotated samples. Table I shows the colors and sample counts for all the classes.

2) Salinas Valley: The SV dataset, acquired by the Airborne Visible Infra-Red Imaging Spectrometer (AVIRIS) over California's Salinas Valley, offers high-resolution imagery (3.7 m) across 512 × 217 pixels. From the original 224 spectral bands, 20 of them affected by water absorption are removed, resulting in 204 usable bands. The region encompasses 16 distinct land cover categories. The sample counts and the color assigned to each class are presented in Table II.

3) Fanglu Tea Farm: The CY dataset was captured using a pushbroom hyperspectral imager over the tea plantation in Fanglu Village, Changzhou, Jiangsu, China. It comprises 80 spectral bands (417–855 nm), with 348 × 512 pixels at 2.25-m spatial resolution. A total of 53 734 labeled pixels are distributed across ten land cover classes. Table III displays the sample counts and colors for all classes.

4) WHU-Hi-LongKou: The LongKou dataset was conducted in Longkou, Hubei, China, using an uncrewed aerial vehicle (UAV)-mounted hyperspectral sensor. It covers a simple agricultural area with six crop types and includes 270 spectral bands (400–1000 nm). The image size is 550 × 400, with a spatial resolution of about 0.46 m. The sample counts and colors for all nine classes are shown in Table IV.

B. Experimental Settings

1) Evaluation Metrics: To assess the model's effectiveness, this experiment adopted three standard evaluation metrics: overall accuracy (OA), average accuracy (AA), and Kappa.

2) Implementation Details: The experiments were conducted using the PyTorch framework. The hardware configuration included an NVIDIA GeForce RTX 4060 Ti GPU, a 64-bit Intel[1] Core[2] i7-14700KF CPU, and 128 GB of RAM. The number of training epochs was set to 50. The Adam optimizer was adopted, and the learning rate was set to 0.0003. To reduce the impact of randomness and enhance the robustness and reliability of the results, all reported outcomes are the average of 20 independent runs.

3) Comparison Methods: To establish the effectiveness of the Conv-J2SMamba approach, we performed a comparative analysis against other advanced methods, including machine learning methods (SVM [12]), CNN-based methods (DSNet [27]), Transformer-based methods [SpectralFormer(Pixel) [32], SpectralFormer(Patch) [32], morphFormer [33]], and Mamba-based methods (3DSS-Mamba [41], S^{2}Mamba [42], and IGroupSS-Mamba [43]) as follows.

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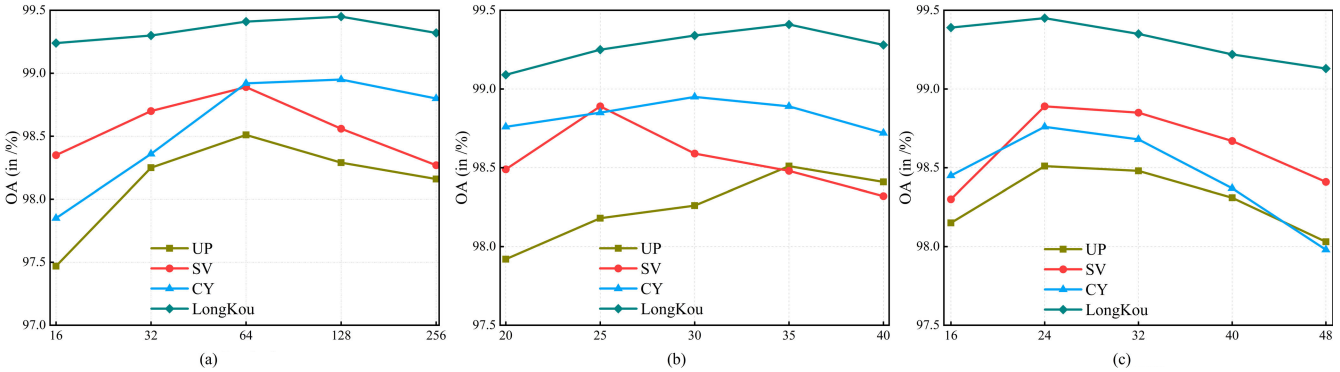


Fig. 5. Analysis of the overall classification performance obtained by our method with different parameters on four datasets. (a) Batch size. (b) Number of principal components retained after applying the PCA transform. (c) Embedding dimension size.

1) *SVM:* The SVM method, utilizing a Gaussian kernel on the libsvm platform, optimizes SVM parameters through fivefold cross validation to address the HSIC problem.
2) *DSNet:* DSNet employs a deep autoencoder unmixing network to automatically integrate convolutional features with subpixel information, further enhancing classification performance.
3) *SpectralFormer(Pixel):* This model consists of group-wise spectral embedding and cross-layer adaptive fusion, which takes each pixel as input to separately learn fine-grained local spectral features and transmit memory-like components across multiple layers.
4) *SpectralFormer(Patch):* Unlike pixel-level approaches, this method captures spatial neighborhood features of the original HSI using patch-level inputs.
5) *MorphFormer:* MorphFormer is a model that utilizes a learnable spectral–spatial morphology network to enhance HSI and CLS token interactions.
6) *3DSS-Mamba:* The 3DSS-Mamba method incorporates a 3-D selective scanning mechanism suitable for HSIC in high-dimensional settings.
7) *Mamba:* S^{2}Mamba uses bidirectional spectral scanning and patch cross scanning to extract spectral and spatial information, along with a fusion gating mechanism to adaptively combine the learned representations.
8) *IGroupSS-Mamba:* This model incorporates interval group spatial–spectral blocks for extracting global spatial–spectral information in multiple directions and scales, organized in groups and layers.

C. Parameter Analysis

1) Analysis of Different Batch Size Values: The batch size refers to the number of training samples used in each iteration of the model. A batch size that is too small may lead to unstable training, while an excessively large batch size can reduce generalization ability and increase the risk of getting trapped in local optima. Therefore, finding an appropriate batch size is crucial. Fig. 5(a) reports the OA achieved by our method under different batch sizes (16, 32, 64, 128, and 256) for the four considered datasets. It is evident that the batch size has a significant impact on classification accuracy. For the UP and SV datasets, the highest accuracy is achieved when the batch size is 64, whereas for the CY and LongKou datasets, the optimal batch size is 128.

2) Analysis of the Number of Principal Components Retained After Applying the PCA Transform: PCA is a widely used dimensionality reduction technique whose primary objective is to simplify the data structure, remove redundant spectral information, and preserve as much original discriminative information as possible. To determine the optimal number of principal components, comparative experiments were conducted with PCA dimensions set to 20, 25, 30, 35, and 40. The classification accuracies under different settings are illustrated in Fig. 5(b). It can be clearly observed that, as the number of principal components increases, the overall accuracy on all four datasets first improves and then declines. This phenomenon occurs because using too few principal components leads to the loss of critical features, whereas an excessively large PCA dimension causes the model to exhibit the Hughes phenomenon—that is, under limited training samples, continuously increasing the feature dimensionality may instead degrade classification performance. For example, the accuracy on the UP dataset begins to decrease when PCA is set to 40. This can be attributed to the relatively limited number of training samples in the UP dataset, making it particularly sensitive to the Hughes phenomenon. Based on the results, the PCA dimensions for the UP, SV, CY, and LongKou datasets are set to 35, 25, 30, and 35, respectively.

3) Analysis of the Size of the Embedding Dimension: To determine the optimal embedding dimension of Conv-J2SMamba, comparative experiments were conducted by systematically varying the embedding dimension, which was set to 16, 24, 32, 40, and 48, corresponding to five different configurations. Fig. 5(c) illustrates the performance trends of these configurations on the four datasets. It can be observed that, as the embedding dimension gradually increases, the model performance first improves and then declines. When the embedding dimension is too small, the feature representation capacity of the model is insufficient; conversely, an excessively large embedding dimension significantly increases the number of model parameters. When the embedding dimension is set to 24, the model achieves the best classification accuracy on all four datasets, striking an optimal balance between feature representation capability and model complexity. Therefore, the embedding dimension is uniformly set to 24 in all experiments.

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TABLE V

CLASSIFICATION PERFORMANCE OF THE SVM, DSNET, SPECTRALFORMER(PIXEL), SPECTRALFORMER(PATCH), MORPHFORMER, 3DSS-MAMBA, S^{2}MAMBA, IGROUPSS-MAMBA AND THE PROPOSED CONV-J2SMAMBA METHODS WITH THE UP DATASET

Category	Classification accuracy achieved by different models (in %)								
	SVM	DSNet	SpectralFormer (pixel)	SpectralFormer (patch)	MorphFormer	3DSS-Mamba	S^{2}Mamba	IGroupSS -Mamba	Proposed
1	92.65	97.50	86.06	91.60	97.95	93.32	**99.02**	98.40	98.05
2	96.88	99.80	96.08	98.68	99.84	97.42	99.87	99.81	**99.91**
3	64.24	87.55	53.53	77.05	92.42	91.02	90.48	93.21	**95.14**
4	86.30	97.08	84.76	94.89	**96.58**	87.05	95.82	90.63	92.31
5	99.14	**100.00**	99.11	99.99	99.88	98.79	99.92	99.10	97.59
6	71.43	98.16	50.89	87.20	99.03	91.83	98.38	99.64	**99.95**
7	34.60	92.64	75.02	65.55	92.21	67.83	84.04	99.51	**99.55**
8	86.47	95.36	86.21	89.08	94.53	90.67	98.61	95.42	**95.93**
9	99.74	**99.60**	99.82	95.90	98.43	84.01	93.93	98.95	96.14
OA	88.17±1.89	97.85±0.28	84.99±0.59	93.02±0.48	98.13±0.44	93.32±1.85	98.08±0.21	98.16±0.07	**98.51±0.07**
AA	81.27±4.18	96.41±0.21	81.28±0.64	88.88±1.06	96.76±0.49	89.10±3.74	95.56±0.51	97.20±0.22	**97.48±0.23**
Kappa	84.06±2.61	97.15±0.37	79.68±0.78	90.70±0.65	97.52±0.59	91.15±2.46	97.45±0.28	97.57±0.10	**98.02±0.05**
Params (M)	—	0.2506	0.1619	0.1711	0.2269	0.0242	0.1204	0.1396	0.0854
Flops (G)	—	0.0082	0.0406	0.0425	0.0447	0.0528	0.0259	0.0199	0.0158
Tr (s)	0.50	68.32	177.99	203.94	113.59	88.24	122.58	164.80	10.76
Te (s)	2.26	8.72	14.10	20.83	56.08	9.21	14.74	42.82	4.51

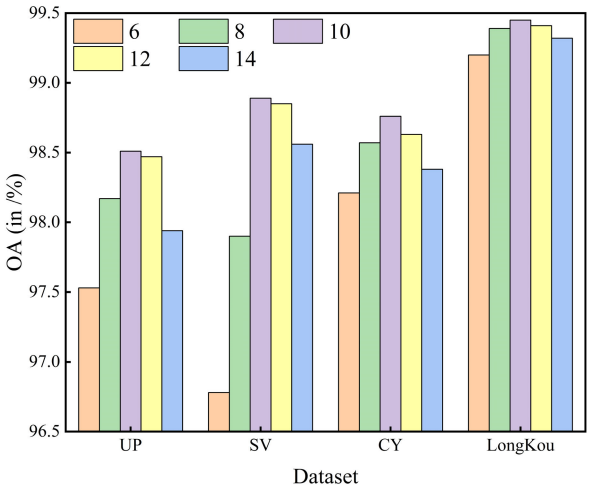


Fig. 6. Sensitivity analysis of different patch sizes on the four datasets.

4) Analysis of Different Patch Sizes: Patch size directly determines the amount of spatial context available to patch-based HSIC models. A very small patch may be insufficient for capturing object-level structures, whereas an excessively large patch size not only introduces redundant neighboring pixel interference but also aggravates the pixel-sharing issue between training and testing samples, thereby affecting the fairness of evaluation. Fig. 6 compares patch sizes of 6, 8, 10, 12, and 14 on the four datasets. The experimental results demonstrate that the proposed method maintains the stable classification performance across a moderate range of patch sizes and generally achieves the best or near-best classification accuracy on all four datasets when the patch size is set to 10. Considering both the sufficiency of spatial contextual information and the controllability of pixel-sharing risk, a patch size of 10 is uniformly adopted in all patch-based experiments in this article.

D. Comparison With State-of-the-Art Methods

A quantitative analysis [in terms of OA(%), AA(%), and Kappa(%)] after applying different methods to the UP, SV, CY, and LongKou datasets is reported in Tables V–VIII, while Figs. 7–10 display some of the obtained classification maps.

1) UP Dataset: In the experiments, the proportion of training samples for the UP dataset was set to 2% of the available labeled samples per class. Table V shows the quantitative classification results obtained by all methods. It is apparent that the proposed approach achieves the highest classification performance. Compared with the second-best method, IGroupSS-Mamba, the OA, AA, and Kappa are improved by 0.35%, 0.28%, and 0.45%, respectively. Moreover, among all the compared methods, the proposed one achieves the highest classification accuracy for five classes, namely, Meadows, Gravel, Bare Soil, Bitumen, and Bricks, while still maintaining competitive superiority on the minority classes 5, 7, and 9. Since SVM struggles to capture the spatial information in the HSI, and the convolutional network used in DSNet has difficulty in capturing global information, their classification performance is limited. SpectralFormer can establish long-range dependencies using the attention mechanism, but its pixel-based approach cannot fully utilize spatial information. Recent Mamba-based methods, such as 3DSS-Mamba, can extract global contextual features without increasing complexity, but perform poorly when the number of samples is small. In contrast, the proposed Conv-J2SMamba exhibits stronger long-range sequence modeling capability. The Joint Scan Merging strategy enables each pixel to acquire a global receptive field, SASSM enhances the ability to capture complex spatial features in the HSI, and CGFF guides the fusion of multilevel features. Conv-J2SMamba comprehensively captures both global and local information and reduces the issue of information loss.

Fig. 7 illustrates some of the classification maps obtained by all the methods used in this experiment. It can be observed that

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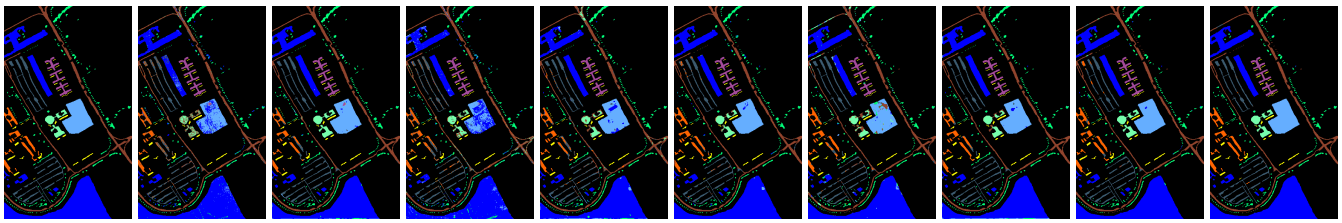


Fig. 7. Classification maps obtained by different methods for the UP dataset using 2% of the available labeled samples per class for training. (a) Ground truth, (b) SVM, OA = 88.17%, (c) DSNet, OA = 97.85%, (d) SpectralFormer(Pixel), OA = 84.99%, (e) SpectralFormer(Patch), OA = 93.02%, (f) MorphFormer, OA = 98.13%, (g) 3DSS-Mamba, OA = 93.32%, (h) S^{2}Mamba, OA = 98.08%, (i) IGroupSS-Mamba, OA = 98.16%, and (j) proposed, OA = 98.51%.

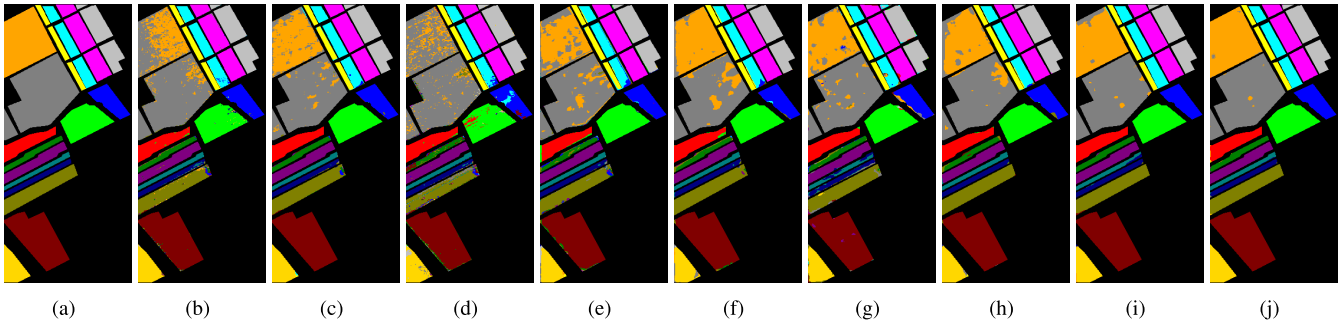


Fig. 8. Classification maps obtained by different methods for the SV dataset using 1% of the available labeled samples per class for training. (a) Ground truth, (b) SVM, OA = 90.12%, (c) DSNet, OA = 95.55%, (d) SpectralFormer(Pixel), OA = 84.27%, (e) SpectralFormer(Patch), OA = 91.08%, (f) MorphFormer, OA = 95.33%, (g) 3DSS-Mamba, OA = 92.67%, (h) S^{2}Mamba, OA = 93.83%, (i) IGroupSS-Mamba, OA = 98.15%, and (j) proposed, OA = 98.89%.

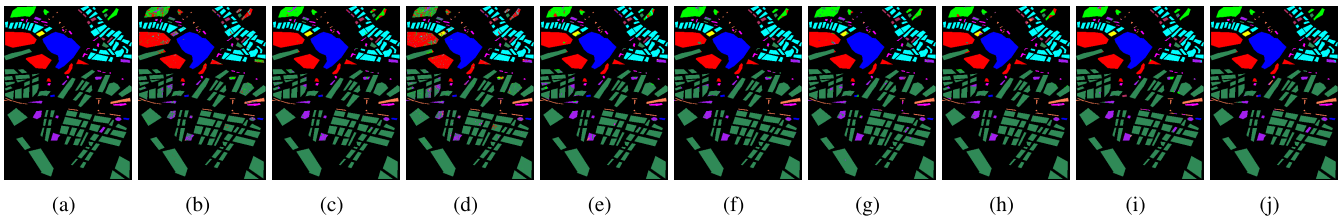


Fig. 9. Classification maps obtained by different methods for the CY dataset using 2% of the available labeled samples per class for training. (a) Ground truth, (b) SVM, OA = 92.25%, (c) DSNet, OA = 98.35%, (d) SpectralFormer(Pixel), OA = 91.77%, (e) SpectralFormer(Patch), OA = 96.73%, (f) MorphFormer, OA = 98.11%, (g) 3DSS-Mamba, OA = 97.42%, (h) S^{2}Mamba, OA = 97.91%, (i) IGroupSS-Mamba, OA = 98.57%, and (j) proposed, OA = 98.95%.

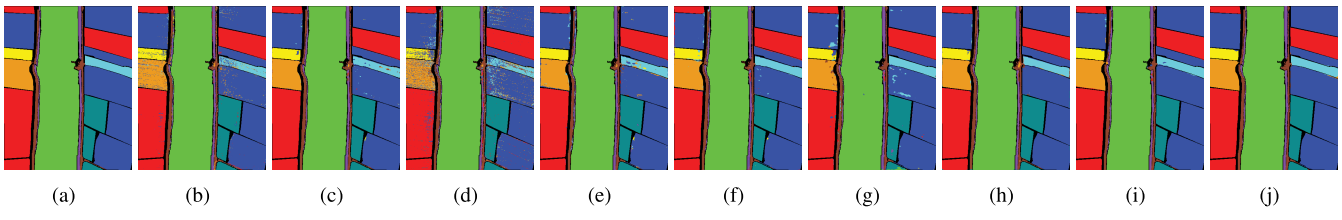


Fig. 10. Classification maps obtained by different methods for the LongKou dataset using 1% of the available labeled samples per class for training. (a) Ground truth, (b) SVM, OA = 97.34%, (c) DSNet, OA = 99.18%, (d) SpectralFormer(Pixel), OA = 94.84%, (e) SpectralFormer(Patch), OA = 98.53%, (f) MorphFormer, OA = 99.01%, (g) 3DSS-Mamba, OA = 98.29%, (h) S^{2}Mamba, OA = 99.25%, (i) IGroupSS-Mamba, OA = 99.26%, and (j) proposed, OA = 99.45%.

SVM and SpectralFormer(Pixel) produce a large amount of noise and exhibit severe confusion in class boundary regions. Although the Mamba-based methods generally outperform the aforementioned approaches, they still suffer from confusion among certain classes. In contrast, the classification maps generated by Conv-J2SMamba contain the least noise and exhibit the smoothest regions. In particular, for classes with complex spatial boundaries, such as Asphalt, Bitumen, and Bricks, the interclass boundaries are delineated more clearly and accurately. These results further demonstrate the strong spatial structure modeling capability of the proposed model.

2) SV Dataset: In our experiments with the SV dataset, we used 1% of the available labeled samples per class for training. Table VI presents the classification results of the nine compared methods. It is evident that Transformer-based classification methods, such as SpectralFormer(Patch), generally achieve lower accuracy than Mamba-based methods. This can be attributed to Mamba's stronger long-range sequence

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TABLE VI

CLASSIFICATION PERFORMANCE OF THE SVM, DSNET, SPECTRALFORMER(PIXEL), SPECTRALFORMER(PATCH), MORPHFORMER, 3DSS-MAMBA, S^{2}MAMBA, IGROUPSS-MAMBA, AND THE PROPOSED CONV-J2SMAMBA METHODS WITH THE SV DATASET

Category	Classification accuracy achieved by different models (in %)								
	SVM	DSNet	SpectralFormer (pixel)	SpectralFormer (patch)	MorphFormer	3DSS-Mamba	S^{2}Mamba	IGroupSS -Mamba	Proposed
1	98.43	99.75	98.08	94.97	97.81	96.63	74.29	**99.99**	99.26
2	98.67	99.37	95.11	98.86	99.79	98.53	99.36	**100.00**	**100.00**
3	95.30	97.79	80.28	94.21	96.80	91.71	91.51	**100.00**	**100.00**
4	99.42	97.88	99.52	94.60	98.43	98.18	98.26	97.77	**99.45**
5	95.54	94.78	93.50	89.58	94.95	95.54	97.25	**99.99**	99.82
6	99.08	100.00	99.28	99.61	**100.00**	99.03	**100.00**	99.96	99.91
7	99.41	99.82	99.06	98.14	99.66	95.01	99.83	99.89	**99.94**
8	91.56	91.46	76.72	85.73	90.11	92.57	90.80	97.88	**97.98**
9	98.98	99.18	96.11	97.85	99.49	97.20	99.97	**100.00**	**100.00**
10	90.85	94.99	83.93	89.25	95.54	87.68	97.76	**99.32**	99.01
11	88.19	92.36	67.19	82.35	93.64	89.02	55.78	**99.40**	98.66
12	99.00	99.86	98.38	99.13	99.54	89.27	99.86	99.91	**99.92**
13	98.57	98.42	95.49	95.76	**99.27**	82.45	98.66	90.44	95.94
14	89.47	96.66	91.70	96.28	99.07	87.07	99.28	95.37	**99.31**
15	54.20	88.76	52.37	77.48	88.68	85.43	88.11	92.45	**96.87**
16	96.05	98.10	83.73	93.38	98.16	87.78	97.06	99.30	**99.63**
OA	90.12±0.22	95.55±0.48	84.27±0.50	91.08±0.94	95.33±0.84	92.67±3.96	93.83±1.76	98.15±0.11	**98.89±0.19**
AA	88.96±0.24	96.83±0.53	88.15±0.69	92.95±1.29	96.93±0.52	92.07±5.13	92.99±3.34	98.24±0.28	**98.88±0.38**
Kappa	93.29±0.45	95.04±0.53	82.46±0.56	90.07±1.04	94.81±0.92	91.83±4.41	93.13±1.96	97.94±0.13	**98.76±0.22**
Params (M)	—	0.4312	0.3564	0.3656	0.2807	0.0243	0.1273	0.1398	0.0853
Flops (G)	—	0.0227	0.1225	0.1263	0.0739	0.0529	0.0275	0.0198	0.0158
Tr (s)	0.37	52.69	232.41	304.83	84.92	52.27	123.12	70.71	6.69
Te (s)	0.84	4.84	18.02	26.29	36.52	12.15	11.18	27.28	5.70

TABLE VII

CLASSIFICATION PERFORMANCE OF THE SVM, DSNET, SPECTRALFORMER(PIXEL), SPECTRALFORMER(PATCH), MORPHFORMER, 3DSS-MAMBA, S^{2}MAMBA, IGROUPSS-MAMBA, AND THE PROPOSED CONV-J2SMAMBA METHODS WITH THE CY DATASET

Category	Classification accuracy achieved by different models (in %)								
	SVM	DSNet	SpectralFormer (pixel)	SpectralFormer (patch)	MorphFormer	3DSS-Mamba	S^{2}Mamba	IGroupSS -Mamba	Proposed
1	94.19	98.55	92.43	97.60	99.30	98.23	99.27	**99.54**	99.39
2	30.39	88.63	16.37	79.95	89.16	84.54	90.76	94.46	**97.87**
3	98.08	99.50	97.85	98.72	99.32	98.86	99.19	99.77	**99.80**
4	9.62	92.10	54.44	94.24	92.10	82.62	74.54	97.36	**99.14**
5	98.56	99.79	98.64	98.72	99.72	99.81	**99.99**	99.98	99.83
6	87.52	92.32	91.79	82.26	**93.81**	92.20	88.83	88.81	88.89
7	86.27	93.74	82.45	91.33	92.10	86.36	75.68	**99.95**	98.55
8	41.16	86.54	44.13	74.88	79.00	84.61	79.35	81.32	**91.90**
9	99.52	99.74	99.63	99.86	99.99	99.17	**100.00**	99.28	98.86
10	90.62	**99.38**	96.13	96.81	98.39	89.88	99.54	93.47	96.07
OA	92.25±0.46	98.35±0.24	91.77±0.32	96.73±0.38	98.11±0.19	97.42±0.55	97.91±0.13	98.57±0.28	**98.95±0.17**
AA	73.59±2.81	95.03±0.69	77.38±1.82	91.44±1.17	94.29±0.95	91.63±1.93	90.72±0.96	95.40±0.77	**97.03±0.61**
KAPPA	88.33±0.73	97.56±0.36	87.61±0.47	95.15±0.56	97.20±0.28	96.17±0.82	96.89±0.19	97.87±0.42	**98.44±0.26**
Params (M)	—	0.2222	0.1349	0.1442	0.1812	0.0242	0.1191	0.1396	0.1355
Flops (G)	—	0.0061	0.0278	0.0293	0.0301	0.0529	0.0256	0.0198	0.0251
Tr (s)	0.44	79.09	185.85	210.54	145.44	104.97	142.39	250.19	11.28
Te (s)	0.94	6.32	13.58	19.45	47.96	11.91	11.84	59.16	3.88

modeling capabilities. Although IGroupSS-Mamba achieves relatively high-overall classification performance, its accuracy in the "Vinyard-untrained" class is suboptimal. In comparison, the proposed method improves the three considered classification metrics by 0.74% (OA), 0.64% (AA), and 0.82% (Kappa), respectively, achieves the highest per-class accuracy in as many as ten classes, and attains perfect classification accuracy of 100% on the Brocoli-green-weeds-2, Fallow, and Soil-vinyard-develop classes. Notably, for Classes 11 and 14, which contain only ten training samples per class, the proposed method still achieves classification accuracies as high as 99%, fully demonstrating its robustness in small-sample scenarios. Fig. 8 shows some of the obtained classification maps. Compared with other methods, the proposed Conv-J2SMamba produces the least noise, and the class confusion in the boundary regions between Grapes-untrained and Vinyard-untrained is also the least noticeable. These results further demonstrate that the proposed method is highly competitive.

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TABLE VIII

CLASSIFICATION PERFORMANCE OF THE SVM, DSNET, SPECTRALFORMER(PIXEL), SPECTRALFORMER(PATCH), MORPHFORMER, 3DSS-MAMBA, S^{2}MAMBA, IGROUPSS-MAMBA, AND THE PROPOSED CONV-J2SMAMBA METHODS WITH THE LONGKOU DATASET

Category	Classification accuracy achieved by different models (in %)								
	SVM	DSNet	SpectralFormer (pixel)	SpectralFormer (patch)	MorphFormer	3DSS-Mamba	S^{2}Mamba	IGroupSS -Mamba	Proposed
1	99.15	99.73	97.67	99.66	99.77	99.67	99.84	99.93	**99.97**
2	89.30	97.80	77.18	95.54	98.65	99.17	98.10	98.92	**99.62**
3	89.38	92.90	65.22	96.01	96.68	95.83	94.58	**98.15**	96.29
4	97.99	99.47	95.47	98.72	99.43	98.05	99.63	**99.81**	99.69
5	76.61	91.09	67.45	87.25	90.57	92.29	93.43	96.23	**99.50**
6	99.51	99.60	98.47	98.95	99.31	98.05	**99.74**	99.58	99.50
7	99.91	99.96	99.96	99.97	99.93	99.63	99.96	99.97	**99.98**
8	90.54	**97.31**	86.85	93.87	94.35	94.19	96.76	91.16	95.70
9	83.26	94.32	72.80	90.66	91.32	85.70	93.16	92.80	**94.43**
OA	97.34±0.21	99.18±0.14	94.84±0.24	98.53±0.15	99.01±0.12	98.29±1.70	99.25±0.05	99.26±0.08	**99.45±0.05**
AA	91.73±1.46	97.04±0.69	84.56±1.21	95.63±0.56	96.67±0.60	95.84±3.66	97.24±0.28	97.40±0.33	**98.12±0.18**
Kappa	96.51±0.29	98.92±0.18	93.20±0.32	98.06±0.20	98.70±0.15	97.76±2.21	99.02±0.07	99.03±0.11	**99.28±0.06**
Params (M)	—	0.5767	0.5489	0.5581	0.3331	0.0242	0.1312	0.1396	0.0854
Flops (G)	—	0.0355	0.1986	0.2036	0.0973	0.0528	0.0285	0.0198	0.0158
Tr (s)	1.77	86.45	431.82	576.40	379.76	203.04	456.17	271.59	17.40
Te (s)	3.66	9.69	55.36	75.86	82.67	45.15	19.63	109.00	11.84

TABLE IX

ABLATION STUDY OF DIFFERENT MODULES IN CONV-J2SMAMBA: SASM AND CGFF

SASSM	CGFF	Overall accuracy for different datasets			
		UP	SV	CY	LongKou
×	×	94.67	97.54	97.34	98.95
√	×	95.46	97.93	97.71	99.21
×	√	96.84	98.31	98.23	99.34
√	√	**98.51**	**98.89**	**98.95**	**99.45**

TABLE X

ABLATION STUDY OF DIFFERENT SCANNING STRATEGIES IN CONV-J2SMAMBA

Scanning strategy	Overall accuracy for different datasets			
	UP	SV	CY	LongKou
SS2D	98.15	98.56	98.48	99.17
ES2D	97.84	98.31	98.13	98.84
Proposed	**98.51**	**98.89**	**98.95**	**99.45**

3) CY Dataset: For the CY dataset, the training sample ratio was set to 2% of the available labeled samples per class. The quantitative classification performance of all evaluated methods is summarized in Table VII. Due to the extremely imbalanced distribution of training samples in the CY dataset, the overall AA values of all methods are relatively low. Nevertheless, Conv-J2SMamba still achieves an AA of 97.03%, surpassing all other comparison methods by at least 1.63%, and attains the highest per-class classification accuracy on four classes. The Reed class (Class 4), which contains only four training samples, is one of the most challenging classes in the CY dataset. For this class, S^{2}Mamba achieves only 74.54% accuracy, while SVM almost completely fails to recognize it. In contrast, Conv-J2SMamba achieves a classification accuracy of 98.33% on the Reed class, demonstrating outstanding generalization capability under small-sample conditions. In the classification maps, as shown in Fig. 9, the other methods suffer from considerable noise, whereas the map produced by Conv-J2SMamba shows only slight confusion in a few boundary regions and remains highly smooth overall.

4) LongKou Dataset: In the case of the LongKou dataset, 1% of the available labeled samples per class was used for training. The classification results of the nine compared methods are presented in Table VIII. As indicated in the table, the suggested approach surpasses all other methods, achieving the best performance across all three metrics. In terms of per-class accuracy, the proposed method attains the highest classification accuracy in five classes and still demonstrates competitive superiority under limited training samples. Fig. 10 further provides the visualization results of all methods. It can be observed that SVM, SpectralFormer, and 3DSS Mamba suffer from significant classification noise, whereas the proposed method exhibits smoother classification regions and less boundary confusion. This further validates the superiority of the proposed approach.

E. Ablation Study

1) Ablation Study on SASSM: SASSM is designed to capture the spatial relationships of adjacent information within the state space. By linearly weighting neighboring states, it aggregates local information into new states, thereby enhancing the ability to model global information. To demonstrate the effectiveness of SASSM, this section compares the classification results obtained by our approach using SASSM and standard SSM while keeping all other parameters unchanged. Table IX presents the overall accuracy across the four considered HSI datasets. The results show that SASSM consistently improves the classification performance on all four datasets, as it establishes neighborhood connectivity and strengthens the flow of contextual information.

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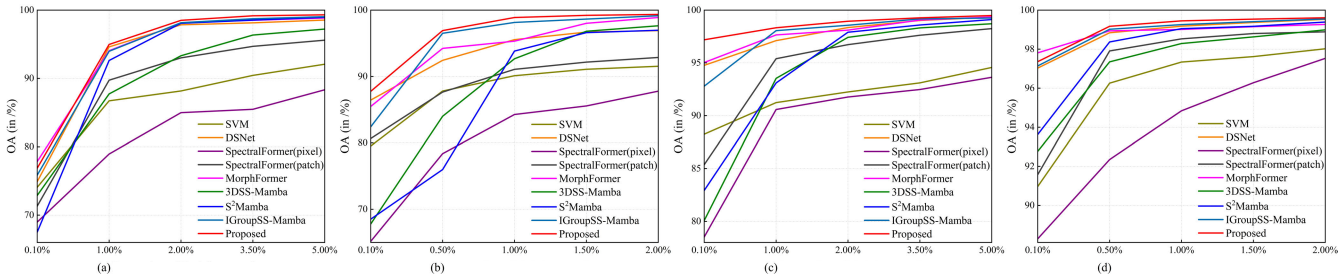


Fig. 11. Classification performance comparison under different proportions of labeled data on four datasets. (a) UP. (b) SV. (c) CY. (d) LongKou.

2) Ablation Study on CGFF: CGFF aims to achieve multilevel feature fusion guided by multiscale attention. It helps the model to extract representative features from global, local, and original features while ignoring less relevant information. To validate the effectiveness of the CGFF module in multilevel feature fusion, a comparative experiment was conducted by implementing our method with CGFF and standard feature fusion, with all other parameters unchanged. This experiment was carried out on the four considered datasets, and the corresponding outcomes are presented in Table X. It is evident that the CGFF module significantly improves classification performance, with accuracy gains of 2.17%, 0.77%, 0.89%, and 0.39% across the four datasets, respectively. This further confirms the effectiveness of the CGFF module in extracting representative features. It is worth noting that the ablation results show CGFF provides a larger performance gain than SASSM. However, it should be emphasized that they address different problems and play complementary roles. SASSM enhances the SSM's ability to capture complex spatial structures of HSIs, mainly improving fine-grained discrimination in regions with complex structures and ambiguous class boundaries. In contrast, CGFF focuses on adaptive fusion of multilevel features and is particularly effective under imbalanced training sample distributions. Together, they jointly support model performance from two perspectives—spatial structure modeling and feature fusion—and are both indispensable.

3) Ablation Study on Different Scanning Strategies: To highlight the effectiveness of the proposed stride-based four-direction scanning strategy, this section compares it with the SS2D and ES2D strategies. Under optimal parameter settings, we evaluated the classification performance of all three methods on four datasets. The results are reported in Table X. The table shows that SS2D achieves the better performance than ES2D. This is because the unidirectional scan of ES2D limits its ability to gather contextual information, whereas SS2D's four-directional scan captures a more comprehensive receptive field. In contrast, our method not only scans in four directions but also employs stride-based scanning to endow every pixel with an omnidirectional global receptive field. The results of our experiments further validate the superiority of the proposed scanning strategy.

F. Sample Sensitivity Analysis

To comprehensively evaluate the generalization ability and robustness of the proposed method under different training conditions, the sample sensitivity analysis is conducted from the following two aspects: 1) different proportions of training samples are selected for each class to investigate the variation trend of classification performance under different sample scales and 2) a spatially disjoint training and testing sample partition strategy is adopted to eliminate the potential influence of pixel sharing on evaluation fairness, thereby providing a more objective assessment of the true classification capability of the model [54].

In the first training scenario, the training sample ratios for the UP and CY datasets are set from 0.1% to 5%, while those for the SV and LongKou datasets are set from 0.1% to 2%. The experimental results of all comparison methods on the four datasets are shown in Fig. 11, where the proposed method is marked with a red line. As can be observed, the classification accuracies of all methods improve to varying degrees as the proportion of training samples increases. Notably, even under the extremely limited training condition of 0.1%, Conv-J2SMamba still demonstrates competitive superiority and consistently outperforms the other methods across most training ratios. This advantage benefits from our dedicated designs for the complex spatial structures of HSIs, including the construction of omnidirectional global receptive fields through the JSM strategy and the explicit modeling of spatial contextual information by SASSM.

In the second training scenario, a spatially disjoint sample partition strategy is adopted, where pixels within a distance of $patchsize//2$ from the center of each training pixel are removed from the testing set. This operation eliminates the spatial overlap between training and testing samples at the patch level, thereby ensuring the fairness of the evaluation process. Table XI presents the classification results on the UP and SV datasets. Compared with the previous classification results, the performance of all methods decreases to some extent. Nevertheless, Conv-J2SMamba still achieves the highest OA (94.57 and 98.05), AA (95.11 and 98.24), and Kappa (93.45 and 97.85) on the UP and SV datasets, respectively. The performance of the proposed method remains within an acceptable range and still surpasses the competing methods. These results indicate that the observed performance gains are not attributed to patch overlap, but rather genuinely reflect the inherent superiority of the proposed method in modeling the spatial structures of HSIs.

G. Complexity Analysis

In this section, we comprehensively compare the number of parameters, FLOPs, training time, and testing time of all methods across four datasets. Tables V–VIII present the

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TABLE XI
ACCURACY OF DIFFERENT METHODS UNDER SPATIALLY DISJOINT TRAINING AND TESTING SETTINGS

Dataset	Metrics	SVM	DSNet	SpectralFormer (pixel)	SpectralFormer (patch)	MorphFormer	3DSS-Mamba	S^2Mamba	IGroupSS-Mamba	Proposed
UP	OA	77.25	92.55	81.51	89.17	94.32	92.42	93.52	93.73	**94.57**
	AA	74.16	92.58	77.41	83.35	93.95	87.36	87.91	92.58	**95.11**
	KAPPA	73.53	92.79	74.94	86.78	93.33	90.81	92.52	92.45	**93.45**
SV	OA	81.39	94.21	83.48	85.29	92.56	90.65	87.02	97.54	**98.05**
	AA	80.65	94.56	87.65	85.74	93.36	91.17	87.89	97.63	**98.24**
	KAPPA	82.41	93.58	81.50	83.63	91.75	89.67	85.59	97.30	**97.85**

experimental results of all evaluated methods. It can be clearly observed that the proposed method achieves nearly the lowest computational complexity while maintaining a favorable balance between computational efficiency and classification accuracy. The following discussion takes the UP dataset as an example. Although SVM requires a relatively short training time, its performance is limited because it ignores the spatial structure information of HSI data. Transformer-based architectures generally exhibit higher parameter counts and longer training times due to their quadratic computational complexity. For example, MorphFormer contains 0.2269M parameters and 0.0447G FLOPs, with a training time of 113.59 s and a testing time of 56.08 s, which are significantly higher than those of the proposed model. Mamba-based methods achieve better computational efficiency than Transformer-based methods because of their linear computational complexity. However, the scanning strategies adopted by existing methods still limit model efficiency. Specifically, the parameter counts of S^2Mamba and IGroupSS-Mamba are 0.1204M and 0.1396M, respectively, both exceeding that of the proposed method. Although 3DSS-Mamba achieves a lower parameter count than our model by adopting a grouping strategy, its performance is inferior. Benefiting from the proposed JSM strategy, the proposed model achieves low computational complexity, with only 0.0854M parameters, 0.0154G FLOPs, a training time of 10.76 s, and a testing time of 4.51 s. Overall, Conv-J2SMamba demonstrates its strong competitiveness and practical applicability in HSIC tasks.

V. CONCLUSION

This article introduces a new Conv-J2SMamba framework for HSIC that effectively integrates global and local feature extraction through convolution and J2SMamba. The concatenation of dual HSIs alleviates feature loss. Benefiting from innovatively constructed stride-based four-directional scanning sequences via the joint scan and merge mechanism, each pixel gains a global and omnidirectional receptive field, significantly enhancing classification performance. The SASSM module greatly improves the competence of the method to capture complex spatial structures in neighborhood features. In addition, to effectively fuse the generated features, the CGFF module leverages multilevel attention to guide the hierarchical feature integration, further boosting the classification accuracy. Experiments on four benchmark HSI datasets demonstrate the superiority of the proposed method.

The proposed method provides a valuable reference for Mamba-based methods in adequately extracting complex spatial structures. Future work will focus on extending this approach to a broader range of remotely sensed data.

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